Run report

2026-10-30T203443Z

Configuration

Parameter	Value	Parameter	Value
batch_size	1,024	n_samples	1,600,000
dim	12	dataset	jets
seed	337,443,673	data_seed	42
hidden_units	128	n_layers	3
n_epochs	100	n_disc_steps	5
lr_g	3.000×10^{-5}	lr_d	1.000×10^{-4}
lambda_dispersion	0.01500	mmd_subsample	16,384
mmd_test	4.237×10^{-4}	best_epoch	91
compile_cache_warm	True		
variables	**Mass and hard scale (IRC-safe kinematics):** m [GeV], $\ln \rho$ **Continuous angularities (IRC-safe jet shapes):** $\lambda^1_{0.5}$, w, λ^1_2 **Splitting and 2-prong substructure:** z_g, $\tau_{21}^{(\beta=1)}$ **Hadronization, multiplicity and fragmentation (IRC-unsafe):** M, n_{ch}, f_{ch}, p_T^D, q		
mmd_sigmas_detector	$3.107 \times (\frac{1}{2},\ \frac{1}{\sqrt{2}},\ 1,\ \sqrt{2},\ 2)$		
mmd_sigmas_particle	$3.027 \times (\frac{1}{2},\ \frac{1}{\sqrt{2}},\ 1,\ \sqrt{2},\ 2)$		

Timing

Phase	Seconds	Share (%)	Detail
DECONVOLVE			
data	2.379	3.7	cache hit – train
train	48.837	76.5	train
transfer	6.251		train
compile	10.258		train
epochs	27.873		train
select	3.839		train
particle_mmd	0.882	1.4	train
save	0.317	0.5	train
evaluate	11.404	17.9	train
total	63.820	100.0	
IBU baseline			
parse_config	0.001	0.0	ibu
data	1.813	14.3	cache hit – ibu
ibu	2.716	21.5	niter=10 – ibu
evaluate	8.113	64.2	12 variables – ibu
total	12.643	100.0	
OMNIFOLD baseline			
parse_config	0.001	0.0	omnifold
data	1.822	0.1	cache hit – omnifold
omnifold	1,736.535	99.4	niter=3, epochs=50 – omnifold
init	3.723		DataLoaders and the two MLPs – omnifold
unfold	1,709.613		MultiFold.Unfold – omnifold
iter1_step1	416.729		detector-level reweighting – omnifold
iter2_step1	409.641		detector-level reweighting – omnifold
iter3_step1	183.714		detector-level reweighting – omnifold
iter1_step2	338.321		particle-level reweighting – omnifold
iter2_step2	237.399		particle-level reweighting – omnifold
iter3_step2	123.773		particle-level reweighting – omnifold
reweight	1.957		evaluating weights on z_target – omnifold
evaluate	8.608	0.5	12 variables – omnifold
total	1,746.966	100.0	
Common			
parse_config	0.002	0.0	plot
data	1.601	2.4	cache hit – plot
load	1.277	1.9	plot
plots	63.553	95.7	plot
total	66.433	100.0	

Detector-level metrics

Distance to Data. Improvements are relative to Simulation.

Wasserstein Distance [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
m [GeV]	208.9	138.8	33.6	**7.048**	**96.6**	10.56	94.9
$\ln \rho$	227.9	20.71	90.9	19.15	91.6	**13.44**	**94.1**
$\lambda^1_{0.5}$	262.6	12.84	95.1	15.94	93.9	**10.80**	**95.9**
w	265.6	**9.978**	**96.2**	14.22	94.6	10.56	96.0
λ^1_2	255.4	**9.215**	**96.4**	12.76	95.0	11.88	95.3
${z_g}^\dagger$	60.45	60.45	0.0	22.82	62.2	**16.73**	**72.3**
$\tau^{(\beta=1)}_{21}$	121.8	70.10	42.4	37.79	69.0	**20.41**	**83.2**
M	224.2	177.3	20.9	**11.42**	**94.9**	22.10	90.1
n_{ch}	220.8	172.3	22.0	**9.265**	**95.8**	23.50	89.4
f_{ch}	52.86	40.94	22.6	6.243	88.2	**5.972**	**88.7**
p_T^D	256.8	30.80	88.0	**12.45**	**95.2**	18.05	93.0
q	79.67	17.51	78.0	5.962	92.5	**4.723**	**94.1**

JS divergence [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
m [GeV]	6.762	3.666	45.8	0.08129	98.8	**0.07650**	**98.9**
$\ln \rho$	8.383	0.4017	95.2	0.1878	97.8	**0.1501**	**98.2**
$\lambda^1_{0.5}$	9.359	0.4714	95.0	0.1415	98.5	**0.1235**	**98.7**
w	9.139	0.4527	95.0	0.1312	98.6	**0.1260**	**98.6**
λ^1_2	7.776	0.1728	97.8	0.1360	98.3	**0.1291**	**98.3**
${z_g}^\dagger$	0.6723	0.6723	0.0	0.1431	78.7	**0.1123**	**83.3**
$\tau^{(\beta=1)}_{21}$	3.372	1.933	42.7	0.3448	89.8	**0.1701**	**95.0**
M	9.640	7.632	20.8	**0.1309**	**98.6**	0.1839	98.1
n_{ch}	7.512	6.314	16.0	**0.1093**	**98.5**	0.1504	98.0
f_{ch}	1.033	0.7960	22.9	**0.08942**	**91.3**	0.09049	91.2
p_T^D	8.223	1.311	84.1	**0.1111**	**98.6**	0.1541	98.1
q	1.576	0.2512	84.1	0.07816	95.0	**0.07768**	**95.1**

VLC divergence [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
m [GeV]	26.93	14.62	45.7	0.3010	98.9	**0.2873**	**98.9**
$\ln \rho$	33.31	1.576	95.3	0.7314	97.8	**0.5805**	**98.3**
$\lambda^1_{0.5}$	37.19	1.877	95.0	0.5603	98.5	**0.4881**	**98.7**
w	36.32	1.800	95.0	0.5171	98.6	**0.4959**	**98.6**
λ^1_2	30.90	0.6583	97.9	0.5043	98.4	**0.4779**	**98.5**
${z_g}^\dagger$	2.683	2.683	0.0	0.5723	78.7	**0.4490**	**83.3**
$\tau^{(\beta=1)}_{21}$	13.44	7.712	42.6	1.374	89.8	**0.6752**	**95.0**
M	38.25	30.31	20.8	**0.4914**	**98.7**	0.7040	98.2
n_{ch}	29.81	25.07	15.9	**0.4094**	**98.6**	0.5746	98.1
f_{ch}	4.125	3.177	23.0	**0.3576**	**91.3**	0.3619	91.2
p_T^D	32.65	5.211	84.0	**0.4403**	**98.7**	0.6117	98.1
q	6.261	0.9940	84.1	0.3025	95.2	**0.3011**	**95.2**

Sliced Wasserstein, joint [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
All	152.4	467.3	−206.5	15.63	89.7	**13.33**	**91.3**

Mean 1D distance over random directions, in Data standard deviations. IBU: product of its per-observable weights.

Particle-level metrics

Distance to Truth. Improvements are relative to Generation.

Wasserstein [$\times 10^3$]

Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
m [GeV]	111.7	94.08	15.8	97.63	12.6	**85.99**	**23.0**
$\ln \rho$	253.1	32.57	87.1	**8.770**	**96.5**	14.63	94.2
$\lambda^1_{0.5}$	303.0	18.71	93.8	13.03	95.7	**6.808**	**97.8**
w	284.1	12.29	95.7	9.636	96.6	**4.957**	**98.3**
λ^1_2	253.8	11.13	95.6	6.942	97.3	**5.165**	**98.0**
z_g [†]	113.4	113.4	0.0	41.21	63.7	**22.35**	**80.3**
$\tau_{21}^{(\beta=1)}$	259.2	150.5	41.9	43.31	83.3	**10.47**	**96.0**
M	256.8	204.8	20.2	33.39	87.0	**21.96**	**91.4**
n_{ch}	234.6	188.6	19.6	37.66	83.9	**23.19**	**90.1**
f_{ch}	45.78	45.33	1.0	8.102	82.3	**3.662**	**92.0**
p_T^D	236.3	55.36	76.6	16.64	93.0	**15.43**	**93.5**
q	79.37	18.15	77.1	7.206	90.9	**3.267**	**95.9**

JS divergence [$\times 10^3$]

Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
m [GeV]	1.963	11.86	−504.2	1.928	1.8	**1.606**	**18.2**
$\ln \rho$	9.878	1.731	82.5	0.1961	98.0	**0.1622**	**98.4**
$\lambda^1_{0.5}$	12.34	2.009	83.7	0.1820	98.5	**0.1704**	**98.6**
w	11.39	1.465	87.1	0.1907	98.3	**0.1853**	**98.4**
λ^1_2	8.824	0.8335	90.6	0.1686	98.1	**0.1249**	**98.6**
z_g [†]	2.041	2.041	0.0	0.3194	84.4	**0.1687**	**91.7**
$\tau_{21}^{(\beta=1)}$	10.50	6.577	37.4	0.5219	95.0	**0.1252**	**98.8**
M	7.842	12.83	−63.7	0.3042	96.1	**0.2441**	**96.9**
n_{ch}	6.447	9.773	−51.6	0.2692	95.8	**0.1786**	**97.2**
f_{ch}	1.025	0.9435	7.9	0.1165	88.6	**0.1116**	**89.1**
p_T^D	8.342	4.029	51.7	0.2669	96.8	**0.2389**	**97.1**
q	2.027	0.9958	50.9	0.1093	94.6	**0.09718**	**95.2**

VLC divergence [$\times 10^3$]

Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
m [GeV]	7.809	45.93	−488.2	7.661	1.9	**6.386**	**18.2**
$\ln \rho$	39.27	6.889	82.5	0.7683	98.0	**0.6335**	**98.4**
$\lambda^1_{0.5}$	48.98	8.014	83.6	0.7221	98.5	**0.6755**	**98.6**
w	45.23	5.837	87.1	0.7434	98.4	**0.7238**	**98.4**
λ^1_2	35.08	3.315	90.5	0.6542	98.1	**0.4827**	**98.6**
z_g [†]	8.086	8.086	0.0	1.276	84.2	**0.6742**	**91.7**
$\tau_{21}^{(\beta=1)}$	41.75	26.17	37.3	2.079	95.0	**0.4943**	**98.8**
M	31.10	49.85	−60.3	1.170	96.2	**0.9340**	**97.0**
n_{ch}	25.58	38.38	−50.1	1.039	95.9	**0.6777**	**97.4**
f_{ch}	4.095	3.766	8.0	0.4654	88.6	**0.4457**	**89.1**
p_T^D	33.05	16.05	51.4	1.055	96.8	**0.9430**	**97.1**
q	7.982	3.921	50.9	0.3830	95.2	**0.3406**	**95.7**

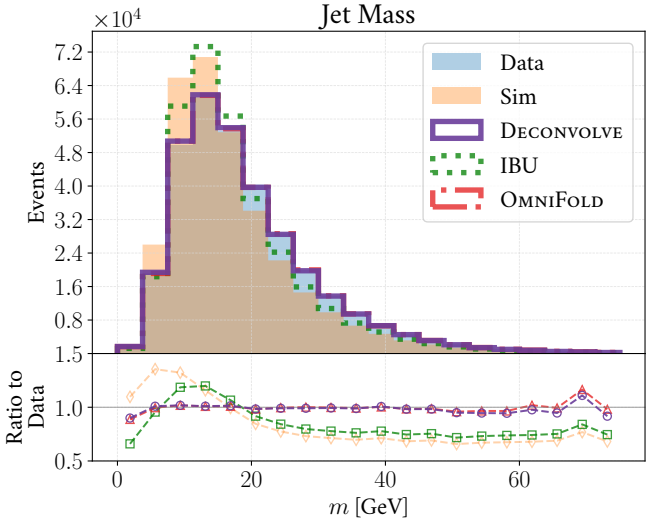
[†] IBU's purity binning produced a single bin, so IBU failed to unfold.

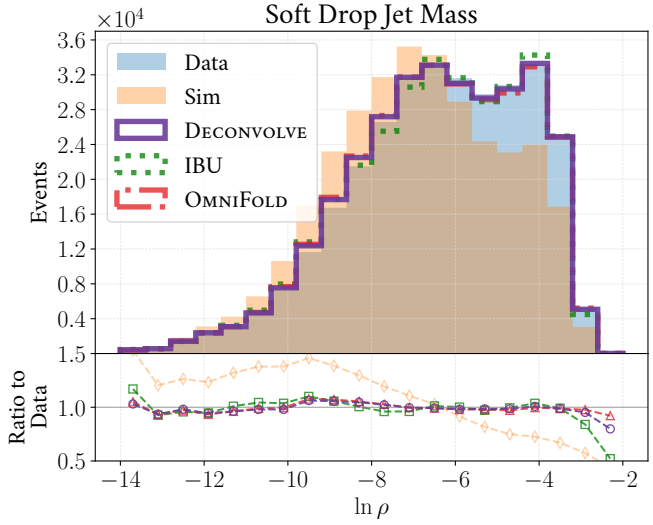
Sliced Wasserstein, joint $[\times 10^3]$

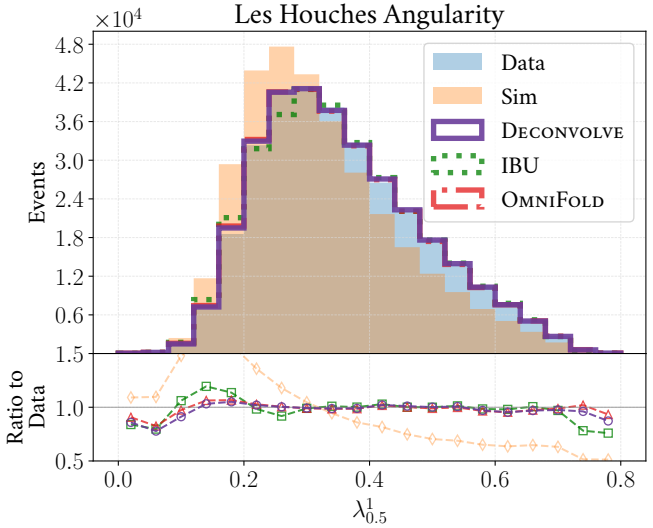
Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
All	174.3	547.6	−214.2	29.37	83.1	**21.27**	**87.8**

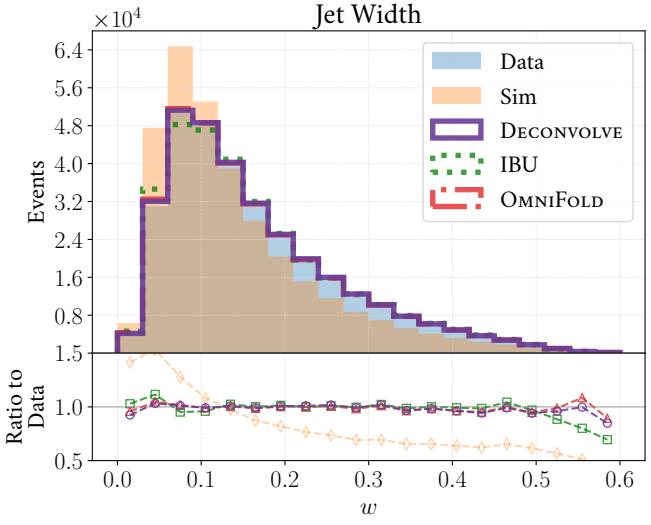
Mean 1D distance over random directions, in Truth standard deviations. IBU: product of its per-observable weights.

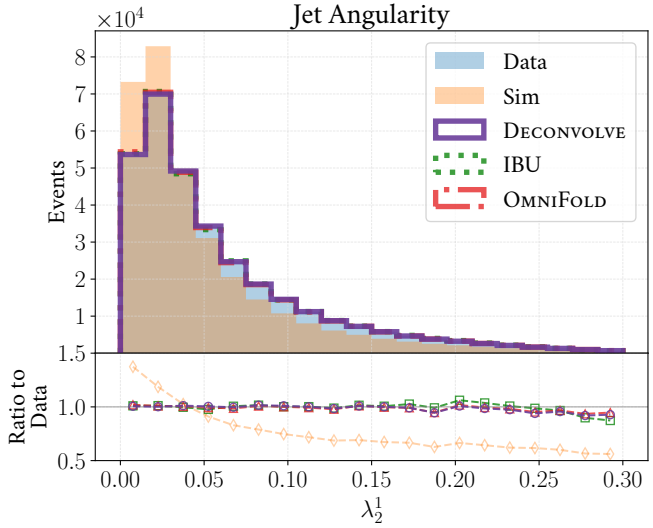
Detector Level (1 of 2)

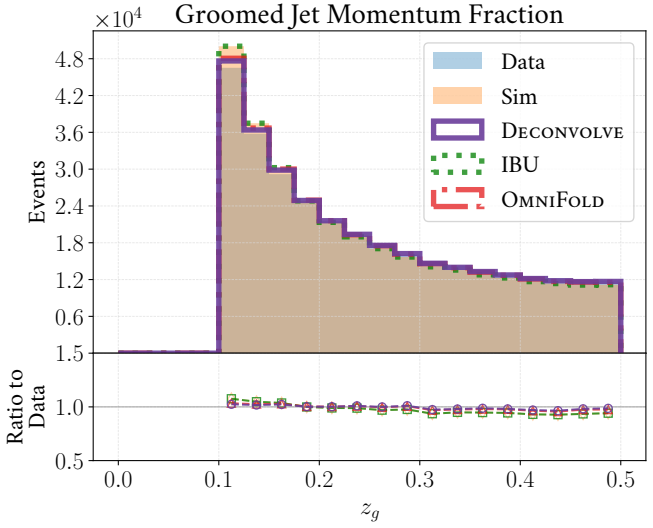




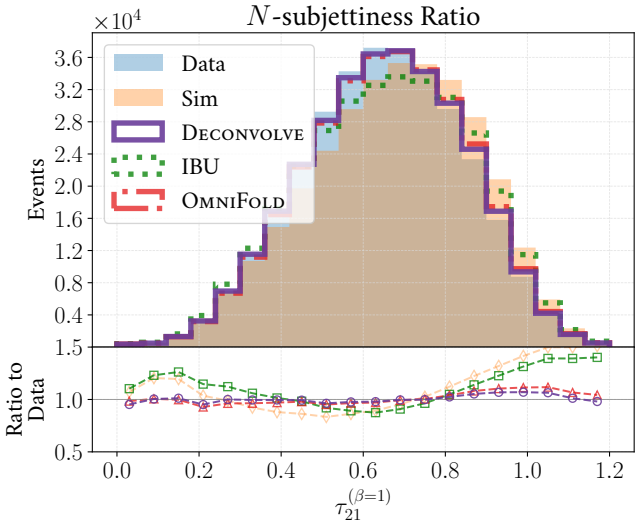


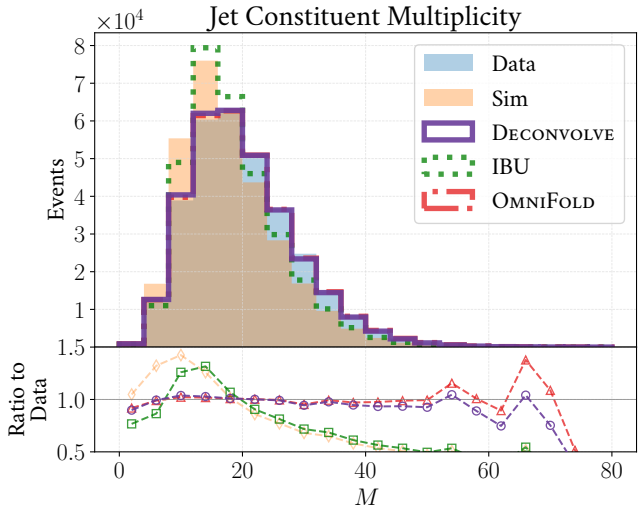


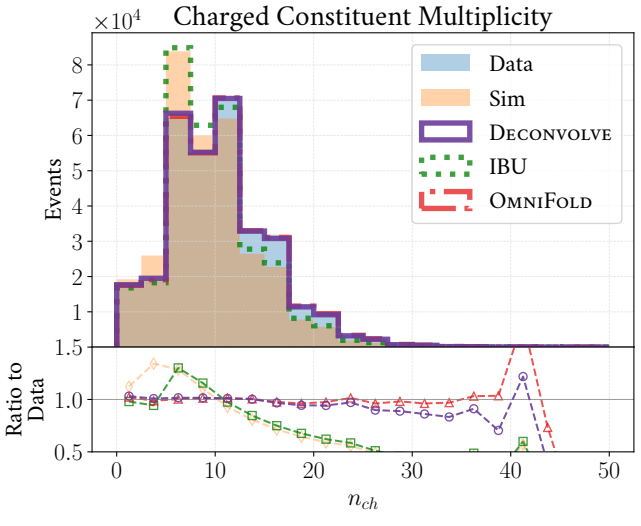


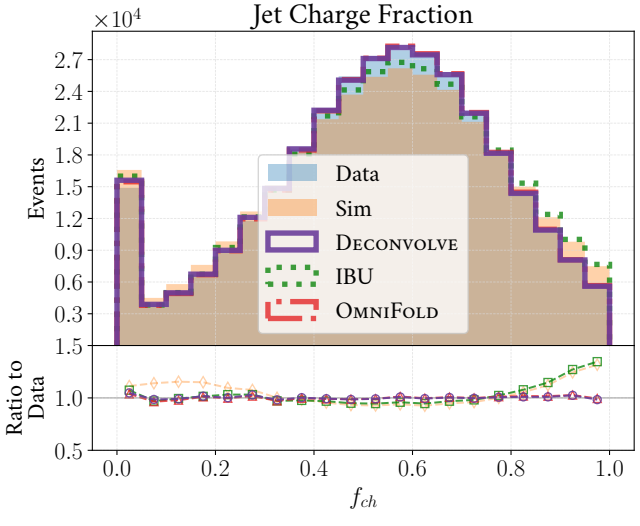


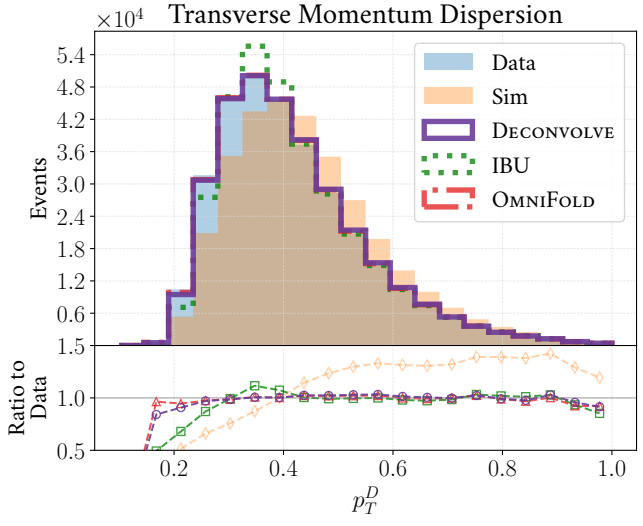
Detector Level (2 of 2)

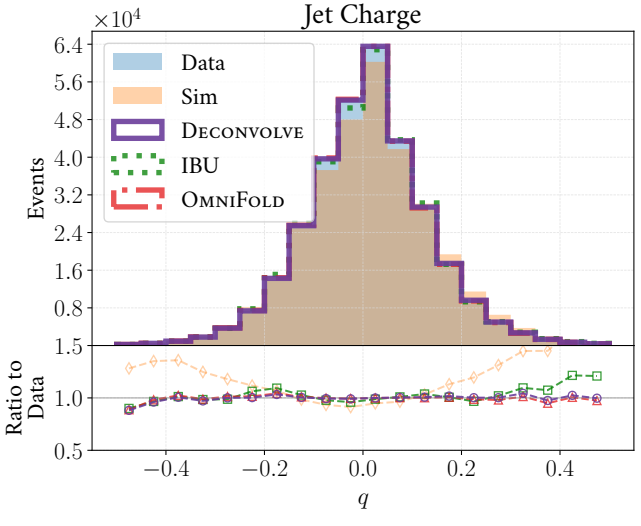




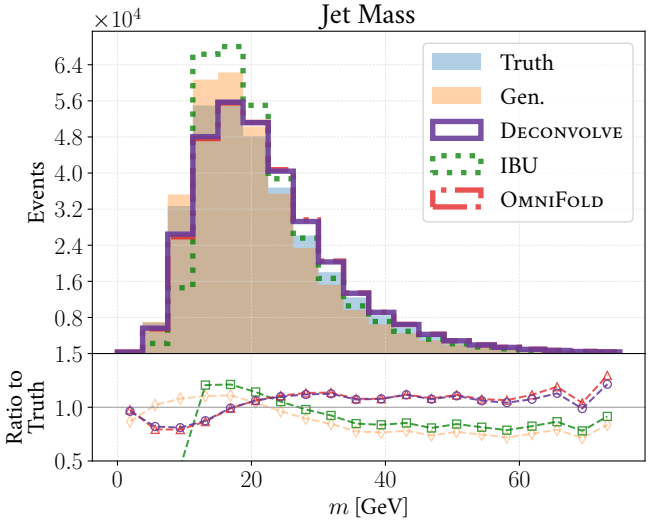


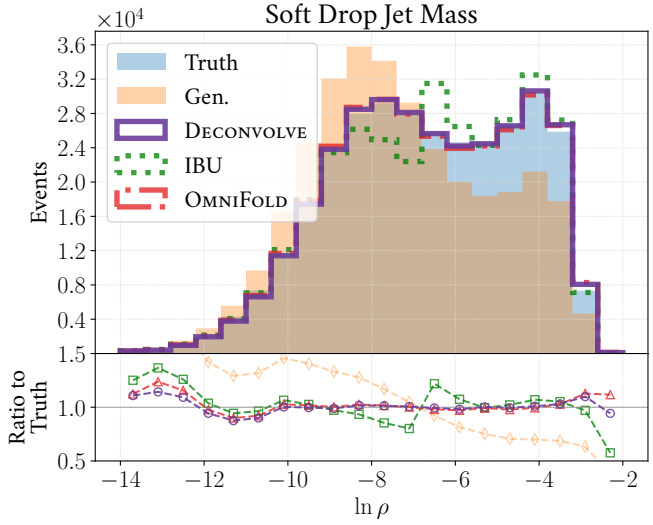


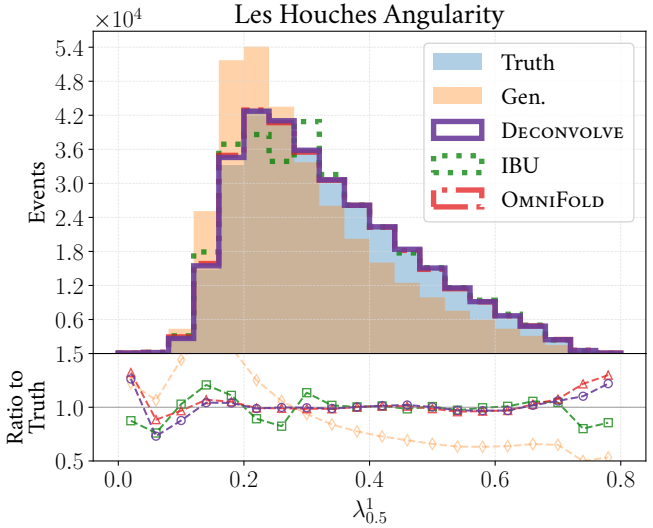


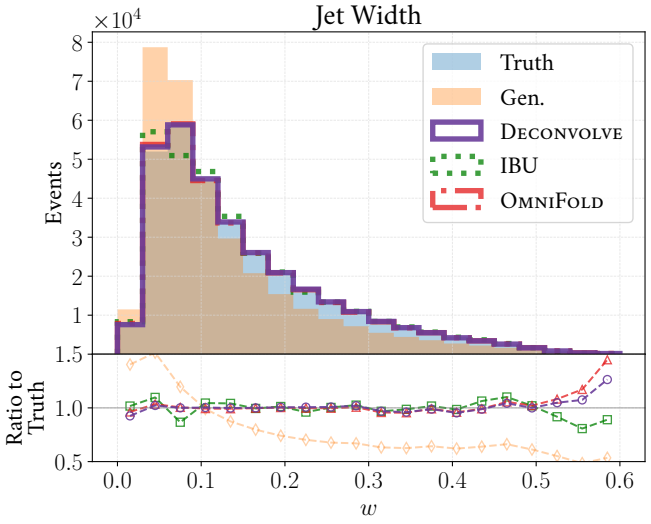


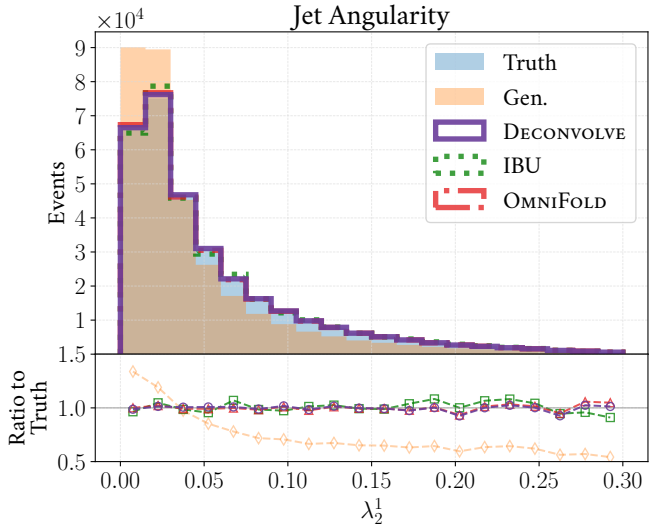
Particle Level (1 of 2)

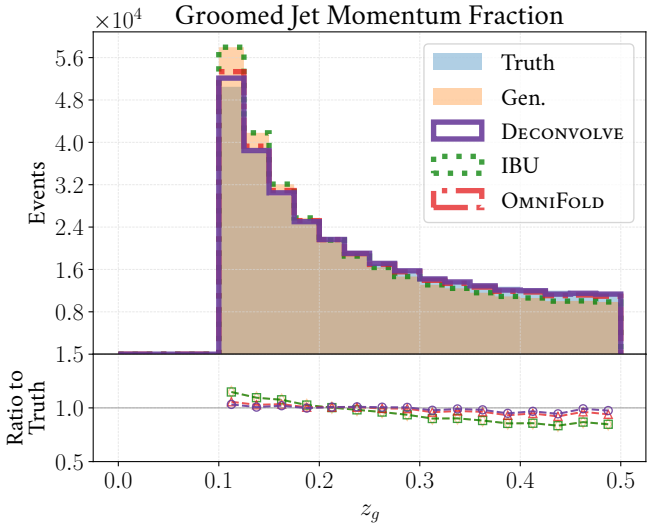




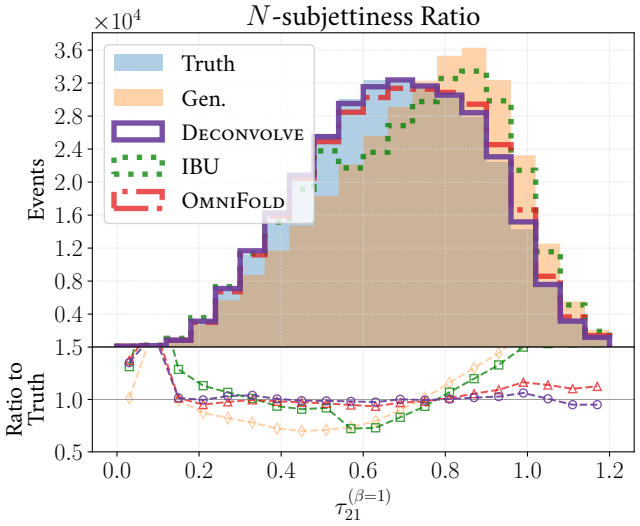


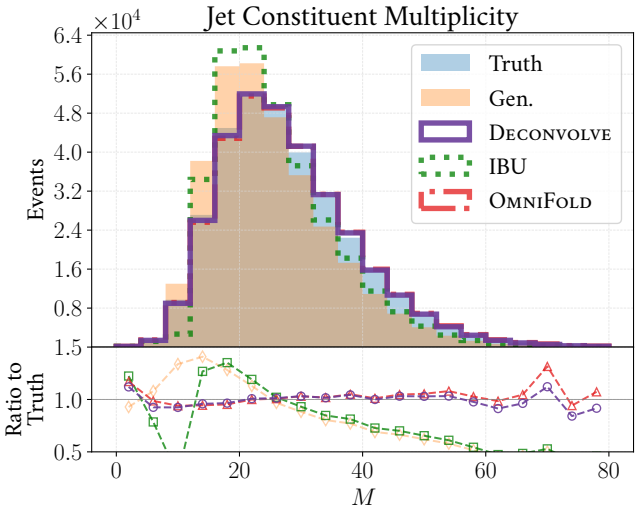


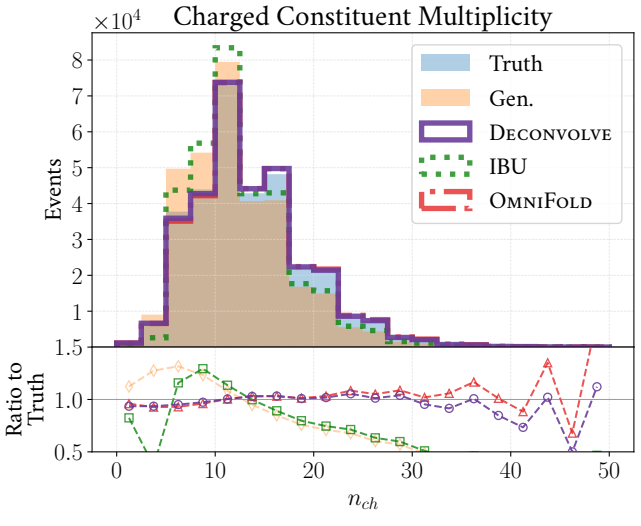


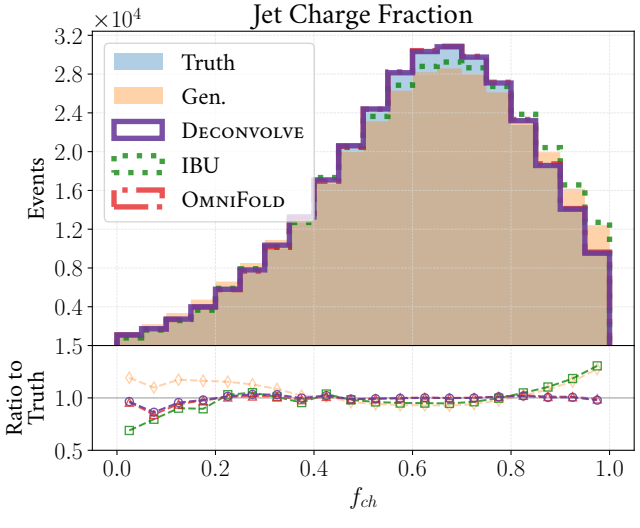


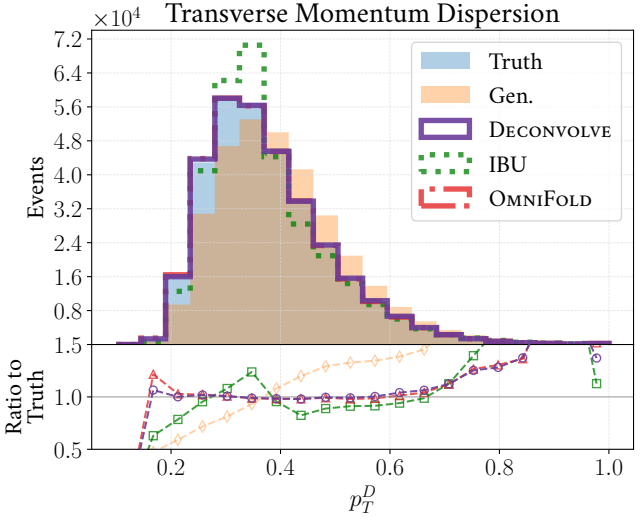
Particle Level (2 of 2)

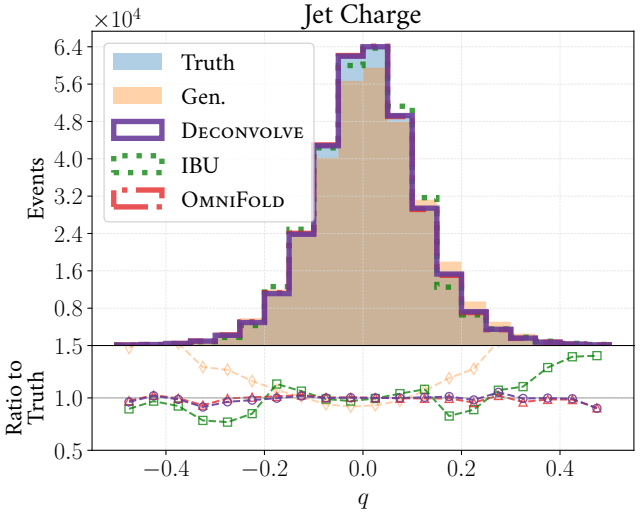












Training diagnostics

Training losses



Selection criteria

